Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Low-Loss RNA Technology for Long-Term Preservation of SHANK3 Autism Patient Organoids			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1 (**Large Category**)**		**Technology Field 2 (**Large Category**)**	
	Bio · Medical		Chemicals	
	Medium Category	Small Category	Medium Category	Small Category
	Bio-processes / Bio-Equipment	Bio-Analytical Instruments	Fine Chemicals	Additives

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Concept of Proposed Technology	Translating biological mechanisms into a human model using patient-derived organoids and engineered tissues is becoming a major area of focus for autism spectrum disorder (ASD) research; in addition to providing insight into developmental and synaptic mechanisms of ASD that cannot be studied in animals alone, organoids and engineered tissues provide insight into how other systems such as the heart and gastrointestinal systems may be affected. Phelan-McDermid Syndrome and SHANK3 Haploinsufficiency represent two of the most well-established monogenetic causes of ASD and both have been shown to affect not only neurons and synapses but cardiac and gastrointestinal systems as well; thus, demonstrating that SHANK3-associated disorders are multi-organ in nature. Organoids derived from patients carrying mutations in SHANK3 and matched controls will provide a powerful tool for studying the effects of SHANK3 mutations on different cell types in a genetically and developmentally relevant human context. While phenotypic assays including morphologic and functional studies, electrophysiology, and gene expression analysis will be used to study the effects of SHANK3 mutations on patient-derived organoids, it is likely that many of these studies will lack the level of mechanistic detail necessary to determine the specific biochemical pathway(s) involved in the AS-related effects seen in the organoids. Therefore, high-throughput RNA-seq will be used to compare the global gene expression of cerebral and cardiac organoids derived from SHANK3 mutation carriers and controls, and to determine if there are commonality in the gene expression changes observed across both tissue types. The use of RNA-seq for the analysis of patient-derived organoids provides several advantages over the use of microarray-based approaches, however, this advantage comes at the cost of significant technical challenges. These include the need for very large amounts of high-quality RNA to produce libraries suitable for high-throughput sequencing, the sensitivity of the transcriptional profile of the cells to changes in culture conditions and handling procedures, and the fact that the transcriptional profile of patient-derived organoids can change dramatically during the course of differentiation and maturation. Because of these limitations, the amount of RNA available from each organoid is usually too small to allow for the production of sufficient numbers of libraries for high-throughput sequencing, and thus, RNA-seq analysis of patient-derived organoids typically requires that RNA extraction and library construction be completed immediately following phenotypic characterization of the organoids. However, in many cases, the practical constraints of phenotyping, freezing and subsequent thawing of multiple patient lines, time points, and treatment conditions, makes it difficult to complete RNA-extraction and library construction immediately after phenotypic characterization of the organoids. To address the current limitations of RNA-seq analysis of patient-derived organoids, we propose the development of an integrated RNA stabilization and low-loss RNA extraction system specifically designed for the analysis of ultra-low-input patient-derived organoids. This system will preserve the transcriptional state of the cells at the point of sampling and will enable long-term storage of the extracted RNA without degradation. The ability to stabilize the transcriptional state of patient-derived organoids post-sampling and to store the resulting RNA for extended periods will greatly facilitate the use of RNA-seq analysis of these cells by decoupling the process of phenotypic characterization from the process of RNA extraction and library construction. Thus, the proposed system will allow researchers to first assess the functional data from the SHANK3 and control organoids- then select the samples of interest and subsequently perform RNA-seq analysis of those selected samples. This new paradigm will improve experimental efficiency, reduce the number of samples requiring sequencing, and enhance data quality by ensuring that RNA-seq analysis is conducted only on samples exhibiting phenotypic differences relevant to the hypothesis being tested. Additionally, the long-term stabilization of the RNA in the patient-derived organoids will also enable reliable storage, transport, and sharing of the organoid samples across multiple sites within collaborative research networks. This capability will be especially valuable for rare genetic diseases like Phelan-McDermid syndrome, where access to patient samples is limited and the ability to collaborate across multiple sites is essential to ensure adequate statistical power for studies of the disease.

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Rationale for Joint R&D Project	To address the need to lock the transcriptional state of ultra-low-input samples upon collection and prevent RNA degradation for long periods of storage, there is a need for effective, pre-analytical stabilization strategies. The stabilization of ultra-low-input samples requires that both RNA integrity and the original gene expression profile be preserved to allow for accurate downstream analysis. Unfortunately, most current preservation solutions have been developed primarily for large tissue sections or cell suspensions and are not optimized for small, 3-Dimensional samples like patient-derived organoids, nor for high-throughput workflows. Furthermore, simply stabilizing RNA is insufficient for ultra-low-input samples. After storage, RNA must be recoverable with minimal loss and compatible with RNA-seq library preparation. This is particularly important when working with rare patient-derived samples (such as cerebral and cardiac organoids derived from SHANK3 mutation carriers and their corresponding controls), which due to their rarity, represent irreplaceable biological materials and thus require special care and consideration regarding sample quantity and preservation. In order to jointly develop an integrated solution that includes preserving the transcriptional state of ultra-low-input samples and efficiently extracting RNA without significant loss, joint research and development with a Korean Small and Medium Enterprise (SME) company focused on developing and improving RNA stabilization and extraction technologies and methods is essential. In this collaboration, the University partners will provide patient-derived organoid models based on autism patients, including cerebral and cardiac organoids derived from SHANK3 and control organoids, to test and improve the new technology in disease relevant conditions. The partnership will leverage the SME's expertise in developing novel reagents and the University's access to well-characterized patient samples and organoid differentiation systems. Ultimately, the developed solution must enable high-throughput processing, minimize the number of manipulations involved in processing samples and produce RNA of sufficient quality and integrity to reliably detect and analyze subtle transcriptomic differences between organoids derived from patient and control individuals.
Key Words	1. Organoid gene expression state locking 2. Minimal-input organoid RNA preservation 3. Low-loss RNA recovery from organoid samples 4. Sequencing-compatible organoid RNA integrity

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3)	**End of the Project**	Level (6)
R&D Objectives	1. Organoid Transcriptional State Preservation: Develop a stabilization reagent that will maintain gene expression profiles at the time of organoid sample collection. Prevent transcriptional drift and artifact creation while the sample is being processed and handled. 2. Long-Term Storage of Organoids Using Ultra-Low-Input RNA Preservative Methods: Allow for long term preservation of ultra-low-input organoid samples to prevent RNA degradation. Preservation chemistry optimization for low-input, small, complex 3D patient derived organoid samples.			

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	3. Low Sample Loss in Organoids through Efficient Extraction of RNA: Ensure the most efficient extraction of RNA from stabilized organoid samples with as little sample loss as possible. Integrate the best stabilization-extraction methods for use on minimal samples with the least amount of sample handling. 4. Preservation of RNA Integrity and Yield for Sequencing: Preserve the quality and quantity of RNA required for preparing RNA-seq libraries from limited amounts of biological materials. Complement low-input sequencing methodologies for analyzing patient derived organoids
R&D Target Specifications	1,Organoid Transcriptional State Preservation: Correlation coefficient ≥ 0.99 on Pearson scale among preserved versus freshly processed organoids Minimal transcriptional bias relative to reference fresh state 2. Ultra-Low-Input Organoid RNA Stabilization: Transcriptional state preservation in ultra-low-input samples for extended storage duration (at least 1 week). RNA fragmentation is no greater than when freshly processed samples are analyzed at extended storage 3. Low-Loss Extraction from Minimal Organoid Samples: Efficient RNA extraction from preserved 3D and microscale organoids. Workflow scalable for HT/batch processing without extra preprocessing/sample handling 4. Organoid Transcriptome Integrity for RNA-seq: RNA quality and yield sufficient for sequencing library prep using minimal biological material. Optimized single workflow that is compatible with RNA-seq without additional preprocessing or transfer steps
R&D Plan	YEAR 1: Development and Feasibility Select a candidate lead stabilization reagent formulation Obtain the baseline transcriptomic profile for SHANK3 and Control Cerebral Organoids Develop an initial integrated stabilization-extraction protocol Preliminary Storage Testing (max. 7 days) Pilot RNA-seq Validation (Target: Correlation coefficient ≥ .95) YEAR 2: Optimization and Validation Maximize RNA Recovery through the Reagent Formulation and Extraction Protocol Extended Validation: Cardiac Organoids and Multiple Patient Lines Long-Term Storage Stability Testing (max. 56 Days) Process Multiple Samples in Batches Validation that Differential Expression can be Detected Between SHANK3 and Controls (Target: Correlation coefficient ≥ .98) YEAR 3: Scale-Up and Application High-Throughput Integrated Workflow Processing (96-Well Format) Finalized Validation of Long-Term Storage (>= 84 Days) and Sample Transport Conditions Standard Operating Procedures and Kit Components Established Multi-Site RNA-Seq Validation Study Completed • Commercialization Readiness Achieved (Target: Correlation coefficient ≥ .99)

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	1. Maintaining Organoid Transcriptional State: Generate SHANK3 patient-derived and matched control cerebral and cardiac organoids. Develop constraints for sample handling to reflect patient-derived organoid workflows. 2. Long-Term Storage of Ultra-Low-Input Organoid RNA: Provision ultra-low-input organoid samples at multiple points in their differentiation process. Determine the storage capabilities of low-input samples for extended periods using rare patient-derived samples. 3. Recovering RNA from Limited Amounts of Organoid Material: Evaluate the effectiveness of RNA recovery using minimal amounts of SHANK3 and control organoid material. Evaluate if the workflow can be used in conjunction with current organoid culture systems. 4. Validating Organoid Transcriptomes to Support RNA-seq Analysis: Compare and evaluate the transcriptomics of preserved organoid samples to those that were stored as fresh samples. Identify potential preservation induced bias and identify disease relevant changes in gene expression
Expected R&R of Korean SME Partner	1. Develop integrated stabilization and low-loss RNA extraction chemistry 2. Optimize protocols and workflows for ultra-low-input and microscale samples 3. Conduct NGS-based technical validation, including RNA integrity, recovery efficiency, and sequencing compatibility 4. Test scalability and robustness of the stabilization–extraction workflow for high-throughput processing

5. Expected Outcomes/Impact

Expected Outcomes and Impact	1. Commercialization-ready stabilization and RNA extraction chemistry kits 2. Reliable transcriptome analysis despite unavoidable processing delays 3. Reduced technical variability in high-throughput organoid, engineered tissue, and MPS studies 4. Support for scalable and reproducible research pipelines using human-relevant biological models

6. References

(1) Salehi and Najafi, RNA Preservation and Stabilization. **Biochem Physiol**, 3:1 (2014).

(2) Sura et al., Pre-Analytical Effects on Whole Transcriptome and Targeted RNA Sequencing Analysis in Cytology: The Effects of Prolonged Time in Storage of Effusion Specimens Prior to Preservation. **Cytopathology**, 34(6) (2023).

(3) Xie et al., The Interference of RNA Preservative and Post-Collection Interval on RNA Integrity from Different Mice Tissues **Genes**, 16, 1421 (2025).

(4) Kathuria, A et al. "Stem cell-derived neurons from autistic individuals with SHANK3 mutation show morphogenetic abnormalities during early development." **Molecular psychiatry** vol. 23,3 (2018): 735-746. doi:10.1038/mp.2017.185

(5) Kathuria, Annie et al. "Transcriptomic Landscape and Functional Characterization of Induced Pluripotent Stem Cell-Derived Cerebral Organoids in Schizophrenia." **JAMA psychiatry** vol. 77,7 (2020): 745-754. doi:10.1001/jamapsychiatry.2020.0196